CHAPTER

10

Fourier Transform Properties

The time and frequency domains are alternative ways of representing signals. The Fourier transform is the mathematical relationship between these two representations. If a signal is modified in one domain, it will also be changed in the other domain, although usually not in the same way. For example, it was shown in the last chapter that *convolving* time domain signals results in their frequency spectra being *multiplied*. Other mathematical operations, such as addition, scaling and shifting, also have a matching operation in the opposite domain. These relationships are called *properties* of the Fourier Transform, how a mathematical change in one domain results in a mathematical change in the other domain.

Linearity of the Fourier Transform

The Fourier Transform is *linear*, that is, it possesses the properties of *homogeneity* and *additivity*. This is true for all four members of the Fourier transform family (Fourier transform, Fourier Series, DFT, and DTFT).

Figure 10-1 provides an example of how homogeneity is a property of the Fourier transform. Figure (a) shows an arbitrary time domain signal, with the corresponding frequency spectrum shown in (b). We will call these two signals: $x[\ ]$ and $X[\ ]$, respectively. *Homogeneity* means that a change in amplitude in one domain produces an identical change in amplitude in the other domain. This should make intuitive sense: when the amplitude of a time domain waveform is changed, the amplitude of the sine and cosine waves making up that waveform must also change by an equal amount.

In mathematical form, if $x[\ ]$ and $X[\ ]$ are a Fourier Transform pair, then $kx[\ ]$ and $kX[\ ]$ are also a Fourier Transform pair, for any constant k. If the frequency domain is represented in *rectangular* notation, $kX[\ ]$ means that both the real part and the imaginary part are multiplied by k. If the frequency domain is represented in *polar* notation, $kX[\ ]$ means that the magnitude is multiplied by k, while the phase remains unchanged.